THE BASE CHANGE ISOMORPHISM CONDITION FOR A SINGLE COMPLEX IS ETALE LOCAL AND DESCENDS TO SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: an etale-local reduction to the scheme case, a stalk computation through strict henselizations, and a faithful flatness argument identifying the comparison arrow as a base change, all inside the delivered span

Editorial note. The label lemma-single-complex-base-change-condition names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The scheme-level base change criterion of Derived Categories of Schemes cited in the statement
- Derived pullback to the small etale site and the strict henselization description of stalks of quasi-coherent complexes on algebraic spaces
- Faithful flatness of the map from a local ring to its strict henselization, and the Tor-comparison lemma for strict henselizations of fibre products

Source reference. Original lemma 08HV: sources/stacks-chow/tags.txt, source lines 10793–10793. Prerequisite pointer invoked at proof line(s) 5476; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 09LQ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 5351–5351. Prerequisite pointer invoked at proof line(s) 5379; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 5410; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07QM: sources/stacks-chow/tags.txt, source lines 9837–9837. Prerequisite pointer invoked at proof line(s) 5452; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DJD: sources/stacks-chow/tags.txt, source lines 16973–16973. Prerequisite pointer invoked at proof line(s) 5473; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DJ8: sources/stacks-chow/tags.txt, source lines 16968–16968. Prerequisite pointer invoked at proof line(s) 5392; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08GH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 896–896. Prerequisite pointer invoked at proof line(s) 5413; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04KF: sources/stacks-chow/tags.txt, source lines 5971–5971. Prerequisite pointer invoked at proof line(s) 5417; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05VP: sources/stacks-chow/tags.txt, source lines 7558–7558. Prerequisite pointer invoked at proof line(s) 5428; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian diagram of algebraic spaces over S. Let $K \in D_{QCoh}(\mathcal{O}_X)$ and let $L(g')^*K \to K'$ be a map in $D_{QCoh}(\mathcal{O}_{X'})$. The following are equivalent

(1) *for any $x' \in X'$ and $i \in \mathbf{Z}$ the map (09LQ) is an isomorphism,*
(2) *for any commutative diagram*

$$\begin{array}{ccccc} & & U & & \\ & & \downarrow & \searrow^{a} & \\ V' & \longrightarrow & V & & X \\ & \searrow^{c} & & \searrow^{b} & \downarrow{\scriptstyle f} \\ & & Y' & \xrightarrow{g} & Y \end{array}$$

with a, b, c étale, U, V, V' schemes, and with $U' = V' \times_V U$ the equivalent conditions of Derived Categories of Schemes, Lemma 0DJ8 hold for $(U \to X)^*K$ and $(U' \to X')^*K'$, and
(3) *there is some diagram as in (2) with $U' \to X'$ surjective.*

Proof. Observe that (1) is étale local on X'. Working through formal implications of what is known, we see that it suffices to prove condition (1) of this lemma is equivalent to condition (1) of Derived Categories of Schemes, Lemma 0DJ8 if X, Y, Y', X' are representable by schemes X_0, Y_0, Y_0', X_0'. Denote f_0, g_0, g_0', f_0' the morphisms between these schemes corresponding to f, g, g', f'. We may assume $K = \epsilon^*K_0$ and $K' = \epsilon^*K_0'$ for some objects $K_0 \in D_{QCoh}(\mathcal{O}_{X_0})$ and $K_0' \in D_{QCoh}(\mathcal{O}_{X_0'})$, see Lemma 071Q. Moreover, the map $Lg^*K \to K'$ is the pullback of a map $L(g_0)^*K_0 \to K_0'$

with notation as in Remark 08GH. Recall that $\mathcal{O}_{X,\overline{x}}$ is the strict henselization of $\mathcal{O}_{X,x}$ (Properties of Spaces, Lemma 04KF) and that we have

$$K_{\overline{x}} = K_{0,x} \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \mathcal{O}_{X,\overline{x}} \quad \text{and} \quad K'_{\overline{x}'} = K'_{0,x'} \otimes^{\mathbf{L}}_{\mathcal{O}_{X',x'}} \mathcal{O}_{X',\overline{x}'}$$

(akin to Properties of Spaces, Lemma 05VP). Consider the commutative diagram

$$\begin{array}{ccc} H^i(K_{\overline{x}} \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y',\overline{y}'}) \otimes_{(\mathcal{O}_{X,\overline{x}} \otimes_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y',\overline{y}'})} \mathcal{O}_{X',\overline{x}'} & \longrightarrow & H^i(K'_{\overline{x}'}) \\ \uparrow & & \uparrow \\ H^i(K_{0,x} \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y',y'}) \otimes_{(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y',y'})} \mathcal{O}_{X',x'} & \longrightarrow & H^i(K'_{0,x'}) \end{array}$$

We have to show that the lower horizontal arrow is an isomorphism if and only if the upper horizontal arrow is an isomorphism. Since $\mathcal{O}_{X',x'} \to \mathcal{O}_{X',\overline{x}'}$ is faithfully flat (More on Algebra, Lemma 07QM) it suffices to show that the top arrow is the base change of the bottom arrow by this map. This follows immediately from the relationships between stalks given above for the objects on the right. For the objects on the left it suffices to show that

$$\begin{aligned} & H^i\left((K_{0,x} \otimes^{\mathbf{L}}_{\mathcal{O}_{X,x}} \mathcal{O}_{X,\overline{x}}) \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y',\overline{y}'}\right) \\ & = H^i(K_{0,x} \otimes^{\mathbf{L}}_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y',y'}) \otimes_{(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y',y'})} (\mathcal{O}_{X,\overline{x}} \otimes_{\mathcal{O}_{Y,\overline{y}}} \mathcal{O}_{Y',\overline{y}'}) \end{aligned}$$

This follows from More on Algebra, Lemma 0DJD. The flatness assumptions of this lemma hold by what was said above as well as Algebra, Lemma 08HV implying that $\mathcal{O}_{X,\overline{x}}$ is the strict henselization of $\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y,\overline{y}}$ and that $\mathcal{O}_{Y',\overline{y}'}$ is the strict henselization of $\mathcal{O}_{Y',y'} \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y,\overline{y}}$. $\square$